Statistical Learning Theory
Exercise Sheet 1

Group 9 - Herzog, Redlich, Kriewald

Part I: Theory

1. A fixed but unknown endpoint

(a) Regression formulation

The feature space and label space are

$$\mathcal{X} = \mathbb{R}^N, \qquad \mathcal{Y} = \mathbb{R}.$$

The joint feature-label space is

$$\mathcal{Z} = \mathcal{X} \times \mathcal{Y} = \mathbb{R}^N \times \mathbb{R}.$$

Since the features

$$U_1, \dots, U_N \overset{\text{i.i.d.}}{\sim} \text{Uniform}[0, L]$$

are independent observations, their joint distribution is the product of the individual distributions:

$$P_U = \text{Uniform}[0, L]^{\otimes N}.$$

Since L is fixed, the label $Y = L$ is deterministic. Its distribution is therefore the Dirac measure

$$P_Y = \delta_L.$$

The joint feature-label distribution $(U, Y) \sim P_L$ is thus

$$P_L = \text{Uniform}[0, L]^{\otimes N} \otimes \delta_L.$$

The squared loss function is defined as

$$\ell(h(u), y) = (h(u) - y)^2, \qquad u \in \mathcal{X},\ y \in \mathcal{Y}.$$

(b) Population risks

The population risk under squared loss is

$$R_L(h) = \mathbb{E}[(h(U) - L)^2].$$

Using the bias-variance decomposition,

$$R_L(h) = \mathrm{Var}(h(U)) + (\mathbb{E}[h(U)] - L)^2.$$

Hypothesis h_1. The first hypothesis is

$$h_1(U) = \frac{2}{N} \sum_{i=1}^{N} U_i.$$

Since $\mathbb{E}[U_i] = L/2$, we obtain

$$\begin{aligned} \mathbb{E}[h_1(U)] &= \frac{2}{N} \sum_{i=1}^{N} \mathbb{E}[U_i] \\ &= \frac{2}{N} \cdot N \frac{L}{2} \\ &= L. \end{aligned}$$

Thus, h_1 is unbiased.

Using independence and $\mathrm{Var}(U_i) = L^2/12$,

$$\begin{aligned} \mathrm{Var}(h_1(U)) &= \frac{4}{N^2} \sum_{i=1}^{N} \mathrm{Var}(U_i) \\ &= \frac{4}{N^2} \cdot N \frac{L^2}{12} \\ &= \frac{L^2}{3N}. \end{aligned}$$

Since the bias is zero, the population risk is

$$\boxed{R_L(h_1) = \frac{L^2}{3N}.}$$

Hypothesis h_2. Let

$$M = \max(U_1, \ldots, U_N).$$

Then $h_2(U) = M$.

For $0 \le t \le L$, the cumulative distribution function of M is

$$\begin{aligned} F_M(t) &= P(M \le t) \\ &= P(U_1 \le t, \ldots, U_N \le t) \\ &= \prod_{i=1}^{N} P(U_i \le t) \\ &= \left(\frac{t}{L}\right)^N. \end{aligned}$$

Differentiating gives the density

$$f_M(t) = \frac{N t^{N-1}}{L^N}, \qquad 0 < t < L.$$

The expectation of M is

$$\begin{aligned} \mathbb{E}[M] &= \int_0^L t f_M(t)\, dt \\ &= \frac{N}{L^N} \int_0^L t^N\, dt \\ &= \frac{N}{L^N} \left[\frac{t^{N+1}}{N+1}\right]_0^L \\ &= \frac{N}{N+1} L. \end{aligned}$$

Similarly, the second moment is

$$\begin{aligned} \mathbb{E}[M^2] &= \int_0^L t^2 f_M(t)\, dt \\ &= \frac{N}{L^N} \int_0^L t^{N+1}\, dt \\ &= \frac{N}{L^N} \left[\frac{t^{N+2}}{N+2}\right]_0^L \\ &= \frac{N}{N+2} L^2. \end{aligned}$$

Therefore, the variance is

$$\begin{aligned}\mathrm{Var}(M) &= \mathbb{E}[M^2] - (\mathbb{E}[M])^2 \\ &= \frac{N}{N+2}L^2 - \frac{N^2}{(N+1)^2}L^2 \\ &= \frac{N(N+1)^2 - N^2(N+2)}{(N+2)(N+1)^2}L^2 \\ &= \frac{NL^2}{(N+2)(N+1)^2}.\end{aligned}$$

The bias of h_2 is

$$\begin{aligned}\mathrm{Bias}(h_2) &= \mathbb{E}[M] - L \\ &= \frac{N}{N+1}L - L \\ &= -\frac{L}{N+1}.\end{aligned}$$

Using the bias-variance decomposition,

$$\begin{aligned}R_L(h_2) &= \mathrm{Var}(M) + \mathrm{Bias}(h_2)^2 \\ &= \frac{NL^2}{(N+2)(N+1)^2} + \frac{L^2}{(N+1)^2} \\ &= \frac{(2N+2)L^2}{(N+2)(N+1)^2} \\ &= \boxed{\frac{2L^2}{(N+1)(N+2)}}.\end{aligned}$$

Hypothesis h_3. The third hypothesis is

$$h_3(U) = \frac{N+1}{N}M.$$

Its expectation is

$$\begin{aligned}\mathbb{E}[h_3(U)] &= \frac{N+1}{N}\mathbb{E}[M] \\ &= \frac{N+1}{N} \cdot \frac{N}{N+1}L \\ &= L.\end{aligned}$$

Thus, h_3 is unbiased.

Its variance is

$$\begin{aligned}\mathrm{Var}(h_3(U)) &= \left(\frac{N+1}{N}\right)^2 \mathrm{Var}(M) \\ &= \frac{(N+1)^2}{N^2} \frac{NL^2}{(N+2)(N+1)^2} \\ &= \frac{L^2}{N(N+2)}.\end{aligned}$$

Since its bias is zero, we obtain

$$\boxed{R_L(h_3) = \frac{L^2}{N(N+2)}.}$$

Comparison for $N = 10$. For $N = 10$, the population risks are

$$R_L(h_1) = \frac{L^2}{30},$$

$$R_L(h_2) = \frac{L^2}{66},$$

and

$$R_L(h_3) = \frac{L^2}{120}.$$

Since $L > 0$, we have

$$\boxed{R_L(h_3) < R_L(h_2) < R_L(h_1).}$$

Therefore, h_3 has the lowest population risk among the three hypotheses for $N = 10$.

Moreover, $R_L(h_1)$ decreases proportionally to $1/N$, while $R_L(h_2)$ and $R_L(h_3)$ decrease proportionally to $1/N^2$. All three risks are proportional to L^2.

2. A two-stage experiment

First draw $L \sim \mathrm{Uniform}[1, 5]$. Given L, draw

$$U_1, \ldots, U_{10} \overset{\text{i.i.d.}}{\sim} \mathrm{Uniform}[0, L].$$

(a) Regression formulation

The feature space and label space are

$$\mathcal{X} = \mathbb{R}^{10}, \qquad \mathcal{Y} = \mathbb{R}.$$

The joint feature-label space is

$$\mathcal{Z} = \mathcal{X} \times \mathcal{Y} = \mathbb{R}^{10} \times \mathbb{R}.$$

Unlike Task 1, the label $Y = L$ is now a random variable with distribution

$$Y = L \sim \text{Uniform}[1, 5].$$

Conditioned on $L = l$, the features satisfy

$$U_1, \ldots, U_{10} \mid L = l \overset{\text{i.i.d.}}{\sim} \text{Uniform}[0, l].$$

Hence, the conditional joint distribution is

$$P_{U|Y=l} = \text{Uniform}[0, l]^{\otimes 10}.$$

The joint distribution P of (U, Y) is characterized by

$$P_Y = \text{Uniform}[1, 5], \qquad P_{U|Y=l} = \text{Uniform}[0, l]^{\otimes 10}.$$

The squared loss function is

$$\ell(h(u), y) = (h(u) - y)^2.$$

Compared with Task 1, the label is no longer deterministic. The distribution P now includes randomness in both the label and the features. The features are conditionally independent given L, but not independent marginally.

(b) Population risks

The population risk of a hypothesis h is

$$R(h) = \mathbb{E}[(h(U) - L)^2].$$

By the law of total expectation,

$$\begin{aligned} R(h) &= \mathbb{E}_L[\mathbb{E}[(h(U) - L)^2 \mid L]] \\ &= \mathbb{E}_L[R_L(h)]. \end{aligned}$$

From Task 1, we know that

$$R_L(h_1) = \frac{L^2}{3N},$$

$$R_L(h_2) = \frac{2L^2}{(N+1)(N+2)},$$

and

$$R_L(h_3) = \frac{L^2}{N(N+2)}.$$

We first calculate $\mathbb{E}[L^2]$. Since $L \sim \text{Uniform}[1,5]$, its density is $f_L(l) = 1/4$ for $l \in [1,5]$.

Therefore,

$$\begin{aligned}
\mathbb{E}[L^2] &= \int_1^5 l^2 f_L(l)\, dl \\
&= \frac{1}{4}\int_1^5 l^2\, dl \\
&= \frac{1}{4}\left[\frac{l^3}{3}\right]_1^5 \\
&= \frac{125-1}{12} \\
&= \frac{31}{3}.
\end{aligned}$$

For $N = 10$, the population risks are:

Hypothesis h_1:

$$\begin{aligned}
R(h_1) &= \mathbb{E}_L\left[\frac{L^2}{30}\right] \\
&= \frac{1}{30}\mathbb{E}[L^2] \\
&= \boxed{\frac{31}{90} \approx 0.3444}.
\end{aligned}$$

Hypothesis h_2:

$$\begin{aligned}
R(h_2) &= \mathbb{E}_L\left[\frac{2L^2}{11\cdot 12}\right] \\
&= \frac{1}{66}\mathbb{E}[L^2] \\
&= \boxed{\frac{31}{198} \approx 0.1566}.
\end{aligned}$$

Hypothesis h_3:

$$\begin{aligned} R(h_3) &= \mathbb{E}_L\left[\frac{L^2}{10 \cdot 12}\right] \\ &= \frac{1}{120}\mathbb{E}[L^2] \\ &= \boxed{\frac{31}{360} \approx 0.0861}. \end{aligned}$$

Thus,

$$\boxed{R(h_3) < R(h_2) < R(h_1).}$$

Among the three hypotheses, h_3 has the lowest population risk, followed by h_2 and h_1.

(c) Empirical risk

Let

$$D_m = \{(U^{(j)}, Y^{(j)})\}_{j=1}^m$$

be a sample of m independently generated labelled examples from the joint distribution P.

Each feature vector consists of ten observations:

$$U^{(j)} = (U_1^{(j)}, \ldots, U_{10}^{(j)}) \in \mathbb{R}^{10}.$$

The empirical risk under squared loss is

$$\boxed{\widehat{R}_m(h) = \frac{1}{m}\sum_{j=1}^{m}\left(h(U^{(j)}) - Y^{(j)}\right)^2.}$$

Here, m is the number of independently generated labelled examples, while $N = 10$ is the number of observations within each feature vector.

Thus, the empirical risk averages over m prediction errors, not over the ten components of an individual feature vector.